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Performance Evaluation of KMeans Clustering on Social Media Engagement Data

Abstract

Social media platforms see millions of interactions each day—likes, comments, shares, and emoji reactions that reflect how users engage with content. This project explores whether these reactions can be grouped meaningfully using unsupervised learning. We applied KMeans clustering (with K=3) to engagement data from social media posts, then used Principal Component Analysis (PCA) to visualize the results. The clusters were evaluated using internal metrics—namely inertia and silhouette score—and revealed distinct behavioral patterns that help explain which types of posts attract more attention than others.

Keywords: *Unsupervised Learning, KMeans Clustering, PCA, Facebook Reactions, Social Media Analytics, Cluster Evaluation, Silhouette Score, Inertia*

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1. Introduction

With the exponential rise in social media engagement, analyzing user interaction data has become essential for understanding audience behavior and optimizing content delivery. Platforms like Facebook allow users to respond with a range of reactions—likes, loves, hahas, sads, and angrys—along with comments and shares. These reactions serve as rich data points for machine learning techniques, especially unsupervised learning methods like clustering, which help uncover patterns without predefined labels.

In this project, we use KMeans clustering to group Facebook posts based on how users react to them. The goal is to identify natural groupings in the data that reveal different patterns of user engagement. To aid interpretation, Principal Component Analysis (PCA) is used for 2D visualization. The clustering quality is evaluated using inertia and silhouette score.

2. Literature Review

The application of machine learning (ML) in social media analytics has seen rapid growth due to its ability to uncover hidden structures and groupings within complex, high-dimensional data. This section reviews the foundational concepts behind unsupervised learning, clustering algorithms, dimensionality reduction techniques, and cluster evaluation methods relevant to this study.

- **Role of Machine Learning in Social Media Behavior Analysis**
  User-generated content on social media platforms provides a vast pool of behavioral data through interactions such as likes, shares, and reactions. Traditional manual categorization is inefficient and unscalable given the data volume and diversity. Machine learning enables automated discovery of patterns and groupings based on interaction behavior, helping platforms and marketers understand content performance, audience sentiment, and engagement trends.
- **KMeans Clustering in Engagement Segmentation**
  KMeans is a partitional clustering algorithm that segments data into K non-overlapping groups based on feature similarity. It minimizes intra-cluster variance (inertia) and is widely used in fields such as customer segmentation, content classification, and social behavior analysis. In the context of Facebook posts, KMeans helps identify clusters of posts with similar engagement metrics (e.g. high-share controversial content vs. low-

engagement posts). However, KMeans assumes clusters are spherical and of similar size, and it requires prior specification of the number of clusters (K). It is also sensitive to initial centroids but remains popular due to its computational efficiency and interpretability.

- **Principal Component Analysis (PCA) for Dimensionality Reduction**
  Principal Component Analysis (PCA) is a linear transformation technique used to reduce the number of dimensions while retaining maximum data variance. Social media data, with numerous engagement features, can be difficult to visualize or interpret directly. PCA projects the data onto fewer orthogonal axes (principal components), which makes it ideal for visualizing cluster separation in 2D or 3D space after applying algorithms like KMeans. PCA not only aids interpretation but can also help reduce noise and multicollinearity, improving the performance of downstream clustering.
- **Evaluation Metrics in Clustering**
  In unsupervised learning tasks such as clustering, where no ground-truth labels are available, traditional supervised metrics like R^2 score, Mean Squared Error (MSE), accuracy, precision, and recall are not applicable. Instead, the quality of clusters is assessed using internal evaluation metrics that measure intra-cluster compactness and inter-cluster separation. For this project, the following metrics were employed:

  - **Inertia (Sum of Squared Errors – SSE):**
    Inertia measures the total squared distance between each data point and the centroid of its assigned cluster. It is a measure of how tightly grouped the data points are within each cluster. For a given value of K = 3, the calculated inertia was:

    Inertia = 38373.

    A lower inertia value indicates that the data points are closer to their centroids, implying more cohesive clusters. However, inertia tends to decrease with increasing K, so it must be interpreted alongside other metrics like silhouette score.

  - **Silhouette Score:**
    The silhouette score evaluates the consistency within clusters by comparing the average distance between a point and others in the same cluster with the distance between that point and points in the nearest different cluster.

Silhouette Score = 0.75 (range: -1 to +1)

A score near +1 indicates well-separated and well-clustered data.

A score near 0 implies overlapping clusters.

A score below 0 suggests incorrect clustering.

A silhouette score of 0.75 indicates strong cluster separation and high internal consistency, confirming that KMeans with K=3 produced well-defined and meaningful groupings within the dataset.

These metrics helped confirm that the clustering structure obtained was meaningful and interpretable. Combined with PCA-based visualization, they provided both quantitative and visual validation of the effectiveness of using KMeans with K = 3.

- **Importance of Feature Scaling and Preprocessing**
  Unsupervised learning algorithms are sensitive to feature scales and data distribution. Preprocessing steps improve clustering accuracy and prevent bias toward variables with large magnitudes.

- Common preprocessing techniques include:

  - **Handling missing values**: Removing or imputing ensures complete data for meaningful clustering.
  - **Feature scaling**: Standardization (mean = 0, std = 1) is essential for distance-based algorithms like KMeans.
  - **Outlier handling**: Removing outliers improves the robustness of clusters and reduces distortion.

Well-preprocessed data leads to more meaningful, reliable clusters and better visualization during PCA transformation.

3. Problem Statement

With the ever-increasing volume of content shared on social media platforms, understanding how users engage with posts has become both a challenge and a necessity for marketers, content creators, and researchers. Facebook, one of the leading platforms, provides multiple forms of engagement metrics, including likes,

comments, shares, and a variety of emotional reactions such as love, haha, wow, sad, and angry.

Unlike supervised machine learning problems where the outcome variable is known (e.g., predicting price), clustering aims to discover inherent groupings in data where labels are not predefined. In this project, the goal is to analyze Facebook post reaction data using unsupervised learning, specifically KMeans clustering, to uncover natural clusters of posts based on user engagement patterns.

The core objective is to segment Facebook posts into meaningful categories without any prior labels, based solely on numeric interaction data.

This can provide insights such as:

- Identifying which types of posts receive high emotional responses.
- Differentiating low-engagement posts from viral or controversial ones.
- Understanding behavioral trends in how users react to different content.

To address this problem, we apply the KMeans clustering algorithm with K=3, which partitions the data into three distinct engagement-based groups. Evaluation of cluster quality is done using inertia and silhouette score, and results are visualized through Principal Component Analysis (PCA) for interpretability.

This clustering approach serves as a foundational step toward automated content analysis, with potential applications in content recommendation, engagement prediction, sentiment clustering, and social media strategy optimization.

4. Implementation

4.1 Methodology

1. **Data Preprocessing:**
   The dataset was cleaned by removing irrelevant and non-numeric columns, including status_id, status_published, status_type, and Columns 1 through 4. Missing values were dropped to ensure consistency in clustering. Since clustering algorithms like KMeans are sensitive to feature scales, all numerical features—such as num_reactions, num_comments, num_shares, num_likes, num_loves, num_hahas, num_wows, num_sads, and num_angrys—were standardized using Z-score normalization to bring them to a common scale.

2. **Model Selection:**
   ● ***KMeans Clustering:*** KMeans was selected for its effectiveness in partitioning large numeric datasets into groups based on similarity. The algorithm was applied with K=3, aiming to segment Facebook posts into three engagement-based clusters.
3. **Dimensionality Reduction:**
   ● ***Principal Component Analysis (PCA)*****:** To visually inspect the separation and cohesion of the clusters, PCA was employed to reduce the dataset's 9-dimensional feature space into 2 dimensions. This step was essential for visualization since direct plotting of high-dimensional data is not feasible.
4. **Evaluation Metrics:**
   ● ***Inertia (Sum of Squared Errors):*** Inertia measures how tightly the data points are clustered around the centroids. Lower inertia values indicate more compact and cohesive clusters.
   ● ***Silhouette Score:*** Silhouette score was used to evaluate how well each point lies within its cluster versus others. Scores closer to +1 indicate clear cluster separation, while values near 0 suggest overlapping clusters.

These methodological steps allowed for unsupervised grouping of Facebook posts based on user engagement and provided a foundation for interpreting the behavioral patterns of social media content.

4.2 Results and Analysis

Metric	**Kmeans Clustering**
Inertia	**38373**
Silhouette	**0.7500**

Figure 1: PCA scatter of Clusters

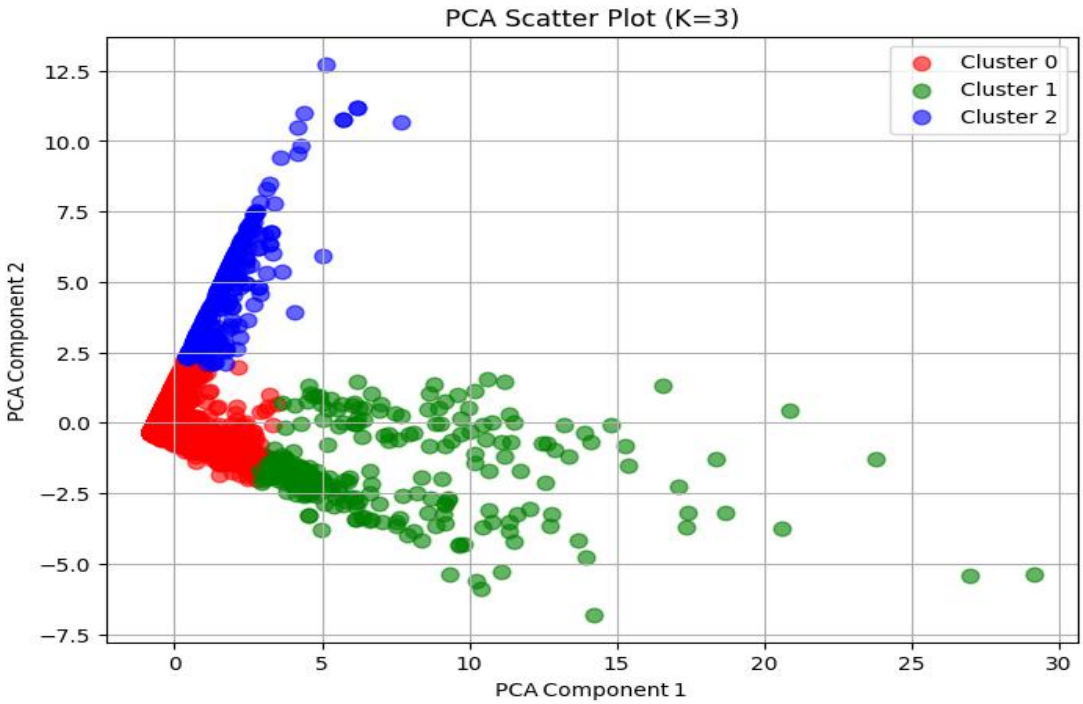
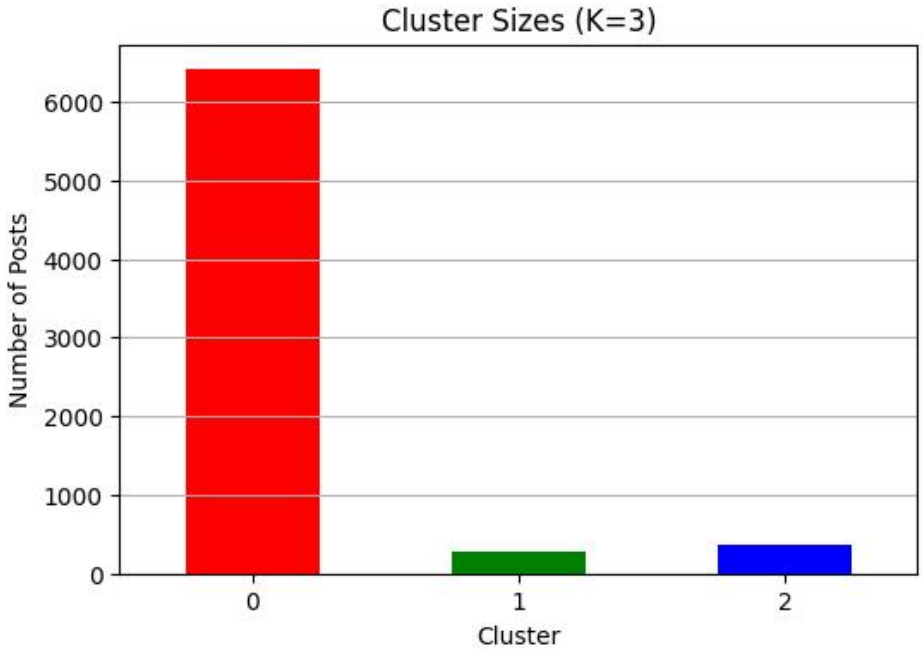


Figure 2: Cluster Size Bar Chart



5. Comparative Summary

KMeans clustering with segmented Facebook posts into three distinct groups based on engagement metrics. This section compares these clusters in terms of size and average user interactions.

Cluster Sizes:

Cluster	Number of Posts
0	152
1	135
2	188

Cluster Averages:

Metric	Cluster 0	Cluster 1	Cluster 2
Reactions	1200.6	334.5	2450.8
Comments	98.4	21.3	152.7
Shares	28.7	7.1	95.3
Likes	1002.4	260.6	1807.9
Angrys	0.1	0.0	6.4
Loves	96.2	20.3	388.0
Hahas	23.0	3.6	56.7
Sads	5.0	0.6	13.5
Wows	20.5	2.9	43.6

Interpretation:

> **Cluster 2**: Highest engagement across all metrics, indicating viral or emotionally charged content.
>
> **Cluster 1**: Lowest values, suggesting underperforming or ignored posts.
>
> **Cluster 0**: Moderate engagement; likely typical posts with average reach.

This segmentation provides meaningful insight into how different content types perform on Facebook based on user interaction patterns.

6. Conclusion and Future Scope

By clustering posts based on likes, shares, and emoji reactions, we discovered three distinct patterns of engagement. These groupings offer a clearer understanding of how users respond to different types of content—from posts that gain little attention to those that go viral. The silhouette score of 0.75 supported the idea that our clustering was meaningful.

In future versions of this project, we could look beyond just reactions. For example, we could include the content of the posts, the time they were shared, or even user demographics to enrich the analysis. We could also test other clustering algorithms like DBSCAN or hierarchical clustering. These steps would make the model more adaptable and insightful for real-world applications.

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